

Suicide Rate Analysis: Relationship of suicide rates to GDP & Age Groups

Introduction

Suicide is a critical global public health issue, taking the lives of 800,000 people each year. September 10th is designated as World Suicide Day every year to draw attention to this neglected and growing mental and public health crisis. For more than two decades, suicide has been a significant public health concern in Asia and Europe, with available data indicating that these regions account for most of the world's suicides. Mental illness is a recognised risk factor for suicide on both continents.

Case Study and Its Importance

The relationship between economic factors and social well-being is a central concern in social sciences. One key aspect of this relationship is the connection between Gross Domestic Product (GDP) *per capita* and suicide rates (Meda et al., 2022). Economic crises, such as the 1997 Asian Financial Crisis, the 2008 Global Financial Crisis, and the European Debt Crisis (2010-2012), have had a major effect on the financial well-being of individuals and communities in both Europe and Asia, leading to fluctuations in GDP per capita (*Asian Financial Crisis | Causes, Effects, & Facts | Britannica Money*, 2023). These fluctuations can be connected to changes in social and psychological stressors, which can subsequently influence suicide rates.

To understand the complicated problem of suicide in Asia and Europe, it is important to analyse suicide rates according to age (Eguchi et al., 2021). The cultures, demography, and economic situations of these two areas are very different from one another. We can gain a better understanding of the intersections between these variables and age-related vulnerabilities by looking at suicide rates among various age groups. This knowledge is essential for developing targeted suicide prevention strategies, mental health support initiatives, and public policies that consider the unique challenges faced by specific age groups within the diverse populations of Asia and Europe. Furthermore, it assists in determining which age groups are more susceptible to economic downturns, enabling governments and organisations to spend resources to protect those who are most at risk (Shah, 2007). In essence, the analysis of suicide rates by age serves as a tool to better understand and address the complex nature of suicide in these regions, ultimately supporting culturally sensitive and effective suicide prevention efforts.

Challenges and Solutions:

Although this case study holds the potential for substantial influence, there are certain challenges to overcome:

1. **Data Quality and availability:** Maintaining data consistency and accuracy across different nations and periods can be difficult. The integrity of the analysis may be impacted by missing or untrustworthy data. However, data quality may be raised by cross-referencing information from several trustworthy sources.
2. **Causation & correlation:** Determining the cause of the association between GDPs per capita and suicide rates can be difficult since several complicated factors could influence the relationship. However, advanced statistical methods can be used to explore the strength and direction of the association while controlling other potential variables.

3. **Ethical Consideration:** Suicide is a sensitive topic with a negative societal reputation. It is important to handle data and findings ethically to avoid sensationalism or causing harm to vulnerable people. Respecting ethical standards and laws when researching suicide will help overcome this challenge.

Data Set

The WHO includes almost all countries represented in the United Nations, but many do not report the injury mortality statistics. Current data were available for 105 countries, which was just above half of all. So, for this analysis, available data were gathered from various sources and combined to get the insights.

Data on rates of death by suicide per 100,000 persons, stratified according to (i) Gross Domestic Product (i) gender (male-female), (ii) age groups (75+, 55-74, 35-54, 25-34, 15-24, 05-14). (iii) Calendar year (from 1985 to 2016), (iv) Country, (v) Human Development Index (HDI), were retrieved from Kaggle and the dataset for 2019 was retrieved from the WHO official website(*Suicide Rates*, n.d.). Gross Domestic Product (GDP) *per capita* and GDP for the year referred to the 2010 US dollar value were downloaded from the World Bank website(*GDP per Capita By Country in Constant 2010 US\$ 1988 - 2021 | WITS Data*, n.d.).

Variable categorization

Variables	Type	Description
Country	String	The Parent dataset has 105 countries
Year	Integer	Details available from 1985-2019
Sex	String	Male & Female
Age	String	75+ 55-74 years 35-54 years 25-34 years 15-24 years 05-14 years
Suicide no	Integer	
Population	Integer	
Suicide/100k Pop	Float	Suicide count per 100,000 people
HDI for year	Float	Human Development Index
GDP for year	String	
GDP per capita	Float	
Generation	String	The Greatest Generation – born 1901-1924. The Silent Generation – born 1925-1945. The Baby Boomer Generation – born 1946-1964 Generation X – born 1965-1979. Millennials – born 1980-1994. Generation Z – born 1995-2012. Gen Alpha – born 2013 – 2025.

		Africa Americas Eastern Mediterranean Europe Southeast Asia Western Pacific
Continent	String	

Table 1: Variables categorization

Dataset Preprocessing

The WHO member countries are grouped into six geographic regions such as Africa, the Americas, the Eastern Mediterranean, Europe, Southeast Asia, and the Western Pacific (*Regional Offices*, n.d.). In this report, the analysis was conducted for all countries and the most characteristic development patterns among those with the highest suicide rates over the past 2.5 decades were chosen and they were categorised as Asia (including the Western Pacific) and Europe (including Russia).

- Asia – republic of Korea, Kazakhstan & Japan
- Europe – Russian Federation, Ukraine, Lithuania

In the process of data preparation for the study, a subset of six countries that exhibited consistently high suicide rates over the past 2.5 decades was selected. Furthermore, as it was the only nation with data for that year, data from 2016 was also omitted. Due to a substantial number of missing data, the human development index (HDI) was eliminated, and the "generation" variable was left out to prevent possible errors in decision-making. To improve the informational value of the data, a new variable representing the continent of each nation was incorporated. There are no known suicide statistics for the age group of 5-14 in 2019. The mean of the previous ten years is added for each nation to prevent missing values.

All data preprocessing and cleaning were carried out using SAS OnDemand, Python & Excel (SAS *OnDemand for Academics*, n.d.).

Statistical Analysis

In this analytical report, we delve into the complex and critical issue of suicide rates in six high-risk countries across Europe and Asia. We have formulated two distinct research questions to comprehensively investigate our case studies.

1. Is there a correlation between a country's GDP per capita and its suicide rates? Does economic prosperity affect suicide rates?
2. How do suicide rates differ among various age groups, and are there any specific age groups that are more vulnerable to suicide?

By addressing these research questions, we aim to contribute valuable insights into the intricate issue of suicide and provide an extensive understanding of its association with economic and demographic factors in these specific regions.

The primary statistical method applied is correlation analysis. This method is used to quantify the strength and direction of the relationship between GDP and suicide rates. Pearson's correlation coefficient is employed to measure this association (*Correlation Analysis | Analyze Correlation between*

Two Variables / QuestionPro, n.d.). Furthermore, regression analysis is used to model the relationship, allowing for the estimation of the impact of GDP while controlling for potential confounding factors.

In order to analyse the second question, Analysis of Variance (ANOVA) is used to evaluate the overall influence of the age categories on suicide rates. This method is well suited since the age variable has several categories in it. Once significant results from the ANOVA test were achieved, Tukey's Honestly Significant Difference (HSD) post hoc test was used. This approach was adopted on purpose as it is specifically designed for situations involving more than two groups, aiming to highlight distinct group distinctions (Frost, 2019). By this, it is possible to identify substantial variations in suicide rates among the age groups.

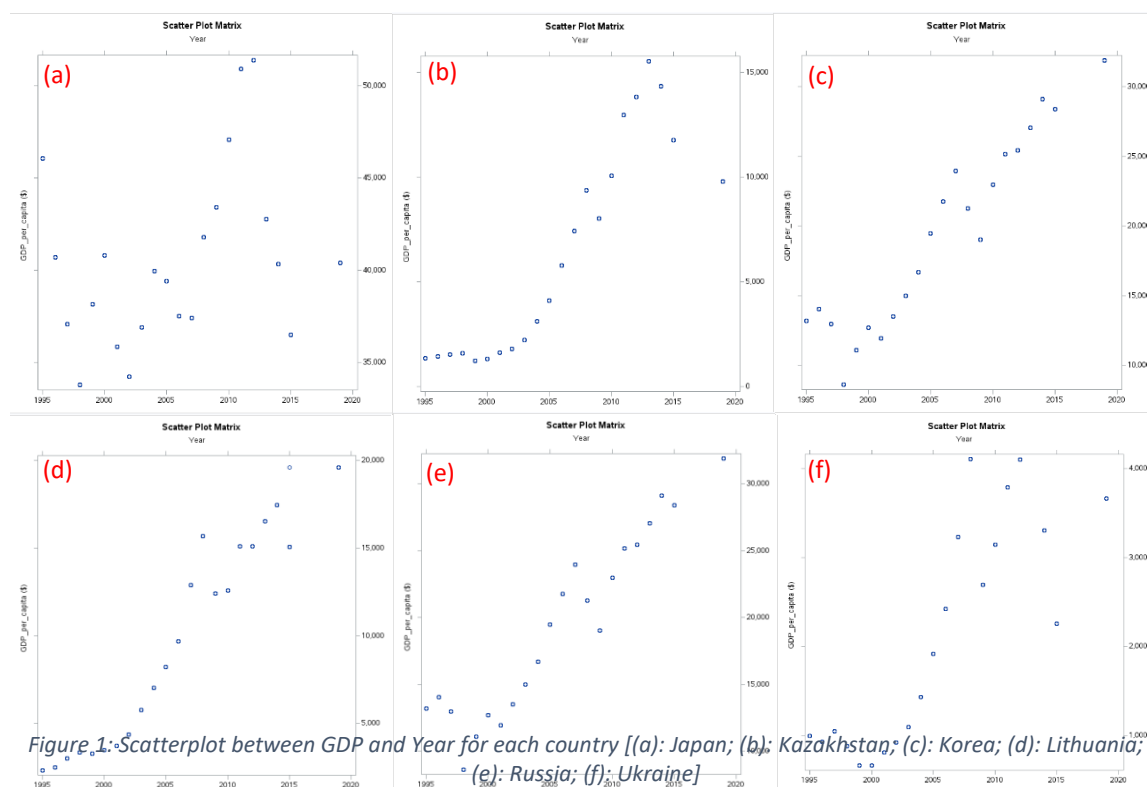
In addition to these statistical methods, visualizations, such as scatterplots, are utilized to provide a clear representation of the data. Scatterplots visually illustrate the relationship between GDP and suicide rates, with each data point representing a specific country. This visual approach aids in identifying patterns, outliers, and trends within the data, enhancing the interpretability of the findings.

Moreover, to enhance the analysis, box plots are used. These box plots will help visualize the distribution of suicide rates with GDP and age groups, shedding light on potential variations and outliers (A Complete Guide to Box Plots, n.d.).

Results & Discussion

Question 1:

In this analysis initially, the correlation between GDPs per capita and the year was examined for the selected 6 countries. Distinct trends were revealed for each country. The figure given below shows the scatter plots between the GDPs per Capita and year for each country. In addition to that person correlation was calculated and given in the table below.



For almost every country, there is a high correlation between year and GDP per capita. i.e., as time goes on, GDP per capita linearly increases. The mean of the calculated parson correlation was 0.8046, indicating a very strong positive linear relationship. This means that looking within the country and asking, “Does an increase in GDP per capita have an effect on the suicide rate” is pretty similar to asking “Does a country's suicide rate increase as time progresses”. Figure 2 given below depicts the suicide

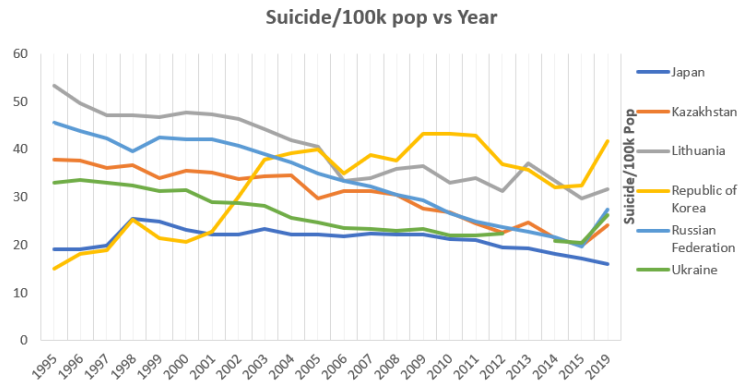


Figure 2: Suicide/100k population vs Year for all six countries

rate trend by year for the selected countries. From the figure given below, we can say that except for the Republic of Korea, other countries have a decreasing trend with time which will give us some hint that the suicide rate and GDP have some negative relationship between them.

Instead of looking at trends within countries, taking all six countries and preparing a scatter plot for GDP per capita and Suicidal rate across all the years will give a better understanding. By using the Linear regression method between Suicide/100k population and GDP per capita we can find the exact relationship.

GDP per capita Vs Year						
Country	Japan	Korea	Kazakhstan	Lithuania	Russian Federation	ukraine
Pearson Correlation	0.33254	0.94409	0.89335	0.96332	0.86052	0.83382
Mean	0.804607					

Table 2: Pearson correlations between year and GDP per capita for each country

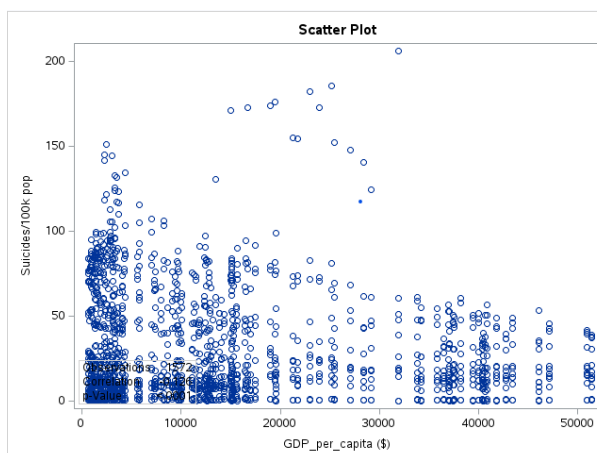


Figure 3: scatter plot of Suicide/100k population against GDP per capita

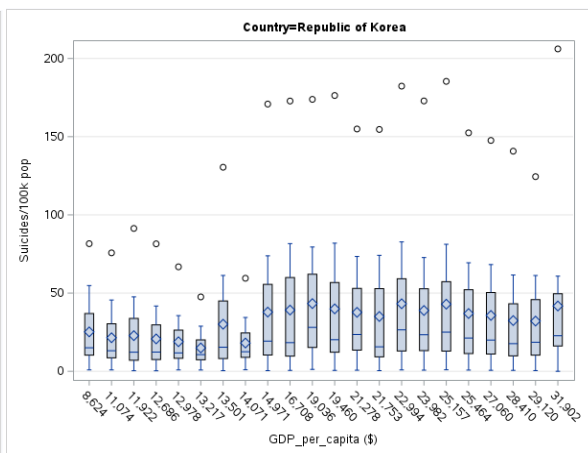


Figure 4: Box Plot of suicide rate against GDP per capita

During the analysis, outliers that could potentially influence a linear regression model were identified. To address this issue, a box plot analysis was employed, with a focus on individual countries.

After analysing the boxplot, it was noticed that a significant number of outliers were present in the data points related to Korea. To make sure that the analysis is reliable, it was decided to remove these specific data entries associated with Korea. This step was taken as the data points were deviating from the overall trend in the data set. This data-cleaning step was undertaken to enhance the reliability of subsequent regression modelling involving suicide rates and GDP per capita.

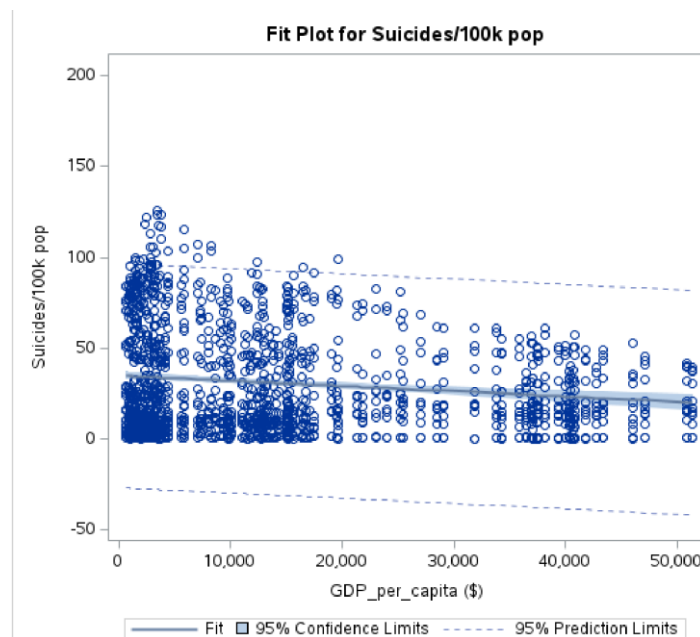


Figure 5: Fit plot for Suicides/100k population against GDP per capita

Root MSE	31.47162	R-Square	0.0159
Dependent Mean	30.73086	Adj R-Sq	0.0153
Coeff Var	102.41049		

Parameter Estimates						
Variable	Label	DF	Parameter Estimate	Standard Error	t Value	Pr > t
Intercept	Intercept	1	34.87500	1.14303	30.51	<.0001
GDP_per_capita (\$)	GDP_per_capita (\$)	1	-0.00028904	0.00005737	-5.04	<.0001

Table 3: Parameter estimates & Root MSE, R-square Adj R-square values.

As per the results in Table 1, the Root MSE is 31.47162, which indicates that, on average, this model's prediction for suicide rates has an error of approximately 31.47 units. The R-square is 0.0159, which is quite low. This means that only about 1.59% of the variance in suicide rates can be explained by changes in GDP per capita. In other words, the relationship between these two variables is weak.

The adjusted R-square is 0.0153, which is very close to the R-square value, indicating that the inclusion of the GDP per capita variable in the model doesn't significantly improve the explanation of the variance in suicide rates. The GDP per capita coefficient is statistically significant ($p < 0.0001$), indicating that it is unlikely to have occurred by chance.

$$\text{Suicide Rate (per 100k population)} = 34.875 - 0.00028904 * \text{GDP per capita (\$)}$$

In this model the coefficient is -0.00028904, indicating that, on average, a \$1 increase in GDP per capita is associated with a decrease of approximately 0.00028904 in the suicide, per 100k population. The negative sign suggests an inverse relation between the variables, but the effect is very small.

The results indicated a weak, inverse association between the two variables. Specifically, an increase in GDP per capita was linked to a small decrease in the suicide rate. However, the model capacity was limited, as evidenced by the low R-Square and Adjusted R-Square values, suggesting that other factors may significantly influence suicide rates.

Question 2:

An ANOVA test was conducted to analyse if there were any significant differences in suicide rates across age categories.

H0: There is no significant difference in suicide rates among different age groups.

Ha: There is a significant difference in suicide rates among at least one pair of age categories.

Dependent Variable: Suicides/100k pop Suicides/100k pop					
Source	DF	Sum of Squares	Mean Square	F Value	Pr > F
Model	5	444635.312	88927.062	122.73	<.0001
Error	1566	1134645.534	724.550		
Corrected Total	1571	1579280.846			

R-Square	Coeff Var	Root MSE	Suicides/100k pop Mean
0.281543	87.56403	26.91747	30.74033

Source	DF	Type I SS	Mean Square	F Value	Pr > F
Age	5	444635.3120	88927.0624	122.73	<.0001

Source	DF	Type III SS	Mean Square	F Value	Pr > F
Age	5	444635.3120	88927.0624	122.73	<.0001

Table 4: table containing overall ANOVA results & Error value tables

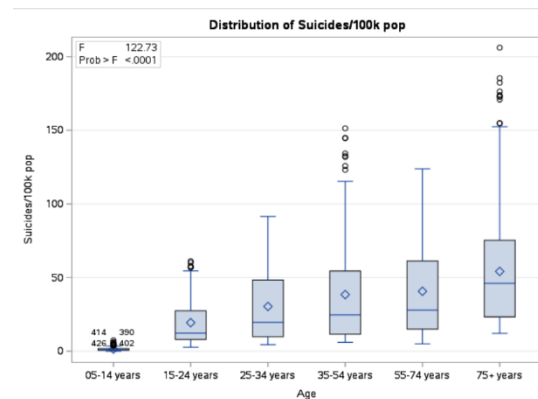


Figure 6: Box plot of suicide/100k population against different age groups

The Analysis of Variance (ANOVA) results reveal a significant influence of age categories on suicide rates. The model's overall performance is statistically significant, as indicated by the very low p-value ($p < 0.0001$). So, the null hypothesis gets rejected. This suggests that age categories play a significant role in explaining the variability in suicide rates.

The model explains approximately 28.15% of the total variation in suicide rates, as indicated by the R-squared value. This moderate R-squared value highlights the relevance of age categories in understanding suicide rate variations.

The Type I SS (sum of Squares) and Type III SS are identical for "Age", and both have an F-value of 122.73 with a very low p-value, indicating that age categories significantly influence suicide rates.

After this, the post hoc test proceeded to determine which specific age groups differ significantly from each other in terms of suicide rates.

Least Squares Means Adjustment for Multiple Comparisons: Tukey		
Age	Suicides/100k pop LSMEAN	LSMEAN Number
05-14 years	1.3856278	1
15-24 years	19.3917176	2
25-34 years	30.3887023	3
35-54 years	38.4386641	4
55-74 years	40.6779771	5
75+ years	54.1592748	6

Least Squares Means for effect Age Pr > t for H0: LSMEAN(i)=LSMEAN(j) Dependent Variable: Suicides/100k pop						
i/j	1	2	3	4	5	6
1		<.0001	<.0001	<.0001	<.0001	<.0001
2	<.0001		<.0001	<.0001	<.0001	<.0001
3	<.0001	<.0001		0.0083	0.0002	<.0001
4	<.0001	<.0001	0.0083		0.9326	<.0001
5	<.0001	<.0001	0.0002	0.9326		<.0001
6	<.0001	<.0001	<.0001	<.0001	<.0001	

Table 4: Least Squares Means Adjustment for Multiple Comparisons: Tukey

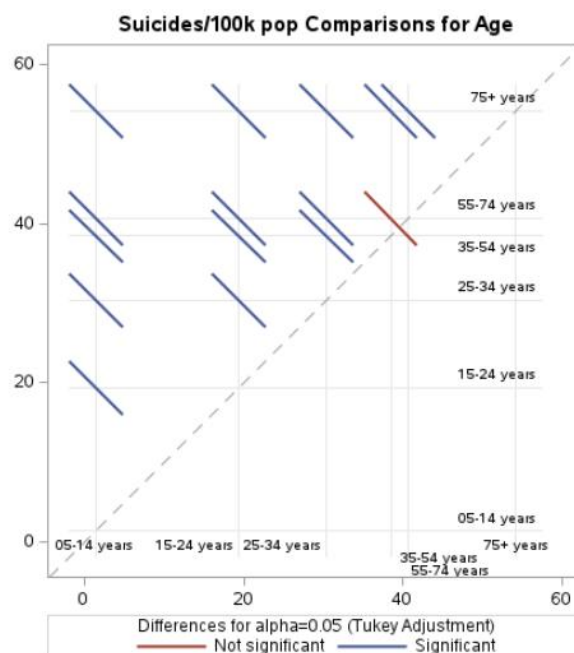


Figure 7: Least Mean Diffogram

To understand the study the Least Squares Means (LSMEAN) values for different age groups and the Tukey adjustment for multiple comparisons were applied to determine significant differences in suicide rates. The analysis revealed that there is a high significant difference between all age groups ($p < 0.001$) except for the comparison between the age categories 25-34 years and 35-54 years ($p = 0.0083$), which shows a moderate level of significance.

Visual representation of Suicide rate according to age groups.

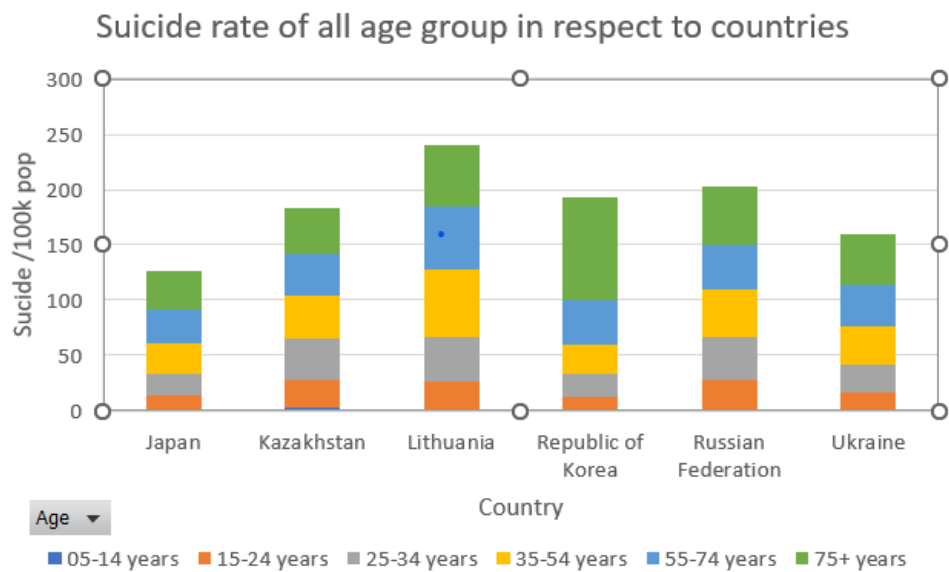


Table 3: Suicide rate of all age group in respect to countries

As per this bar chart it is evident that the result, we got in ANOVA test is almost true, there is a significant difference between suicide rate per age groups.

Conclusion

In conclusion, this study clarifies the complex link that exists between age, economic conditions, and suicide rates in particular Asian and European regions. It explains that although GDP per capita and suicide rate have a small negative relationship, other factors likely have a more significant impact. Moreover, age has a significant role in suicide rate variations, with age groups showing clear disparities in suicide rates. These results highlight the necessity for a comprehensive suicide prevention plan that takes demographic and economic factors into account. By addressing the complications, we can work towards more effective and targeted interventions to combat the global public health crisis of suicide.

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